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(1) Research Program/Project/Study Title: Physico-Chemical Assessments of the Riverine in Roxas City under the RISE Capiz R&D Program (Riverine Improvement towards a Sustainable Ecotourism in Roxas City, Capiz, Philippines)	
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Program or Project has component (if no proceed to 2)? ☒ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food
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	Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation

Municipality/City:	Roxas City
Province:	Capiz
Region:	VI

(5) Classification

Research: ☒ Basic ☐ Applied	Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity

Education	(8) Discipline/Field of Specialization Natural/Biological Science
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(9A) Abstract

This research was conducted to assess water quality using the Physico-chemical parameters of Sitio Cablatan, Brgy. Cagay and Lawis Bay-Bay, Roxas City, Capiz. Water samples were collected from the surface and 3-12 ft depth from nine selected stations in the Palina River and Cadimahan River to achieve this purpose. The physicochemical parameters, such as the water temperature, pH, dissolved oxygen, conductivity, and salinity, were analyzed using the water samples. By assessing the Physico-chemical data according to the National Water Resources Board (NWRB) water pollution control regulation of the Philippines, the water quality classes of the chosen stations in the Palina River and Cadimahan River were determined. The water temperature, conductivity, and salinity of the Palina River and Cadimahan River were within desirable limits. Based on the values of the obtained Physico-chemical parameters, it can be concluded that the Palina River and Cadimahan River water quality was good and did not show any significant pollution problem. The results obtained from the present study shall be helpful in the future management of the same study.

(9B) Keywords

Physico-chemical, salinity, pollution, conductivity, water temperature, pH, dissolved

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oxygen

(10) Introduction

The riverine communities within Capiz are on the frontlines regarding the province's tourism, beauty, economy, and livelihood. It has an abundance of food and recreational activities for tourists, and it has the potential to become the most beautiful site in Capiz. These communities have striking and luscious mangroves that give shelter to aquatic animals. These factors are significant aspects of our riverine communities that can attract tourism.

Our riverine communities have a lot to offer. Protecting and actively improving the beauty of our riverine communities must be a priority. Rivers are a source of income and heritage for the next generation. Tourism is an integral part of the economy. It plays a significant role in developing new infrastructure, hiring employees, and, most importantly, tourists will be exposed to the culture.

Tourism is a significant economic activity that has developed significantly over the years. It's an activity recognized in both developed and developing nations. In universal aspects, tourism is the movement of a person to a country or place outside their usual setting for business or personal purposes. Moreover, tourism can boost the economy, generate more job opportunities, develop infrastructure in the community, and give people a chance to understand the culture and religious aspects of the country.

Water is considered the main requirement for human and industrial growth. Over the years, the increase in industrialization and population was palpable, and the demand for freshwater has also increased. This demand was fulfilled by rivers that provide water for human and agricultural purposes. Due to the waste discharged from human and industrial activities, river water quality has deteriorated, affecting human and aquatic life. Water quality is too impoverished for human consumption due to pollutants contamination.

According to Arain et al., (2008), water is an essential component for the survival of life on earth; it comprises crucial minerals for humans and aquatic animals. Rivers, lakes, and water reservoirs are the planet's essential freshwater resources that provide innumerable benefits. Rivers are used for irrigation purposes, and it provides an ecosystem for aquatic life. They have immense significant elements in the world's biological diversity. They have substantial social and economic benefits due to tourism and recreation and are culturally and aesthetically crucial for people worldwide.

Water is continually considered a requirement for human and industrial growth in every aspect. The increase in population and industrialization also increases the demand for freshwater. This demand is being fulfilled by our rivers that provide water for human and agricultural purposes. However, due to the upsurge in population and industrial activities, the river quality has deteriorated over the years, affecting humans and aquatic life.

The physicochemical parameters are vital and significant to test the water before it can be used as drinking water, for agricultural or industrial purposes, and for domestic. It should be tested with different physicochemical parameters. The selection of the tested parameters depends on water usage and to what extent we need its quality and purity. It is

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essential to get the exact idea of the quality of water and then compare it with standard values. According to Patil *et al.*, 2012, an analysis of water quality is essential to protect the natural ecosystem.

The study area was conducted in the two riverine communities of Roxas City, the Palina River, situated at Sitio Cablatan Brgy. Cagay, and Cadimahan River, situated at Lawis Bay-Bay Rixas City. The two riverine communities were investigated for their physicochemical characteristics, such as water temperature, conductivity, pH, dissolved oxygen, and salinity. This study was carried out to assess water quality using physicochemical parameters.

Generally, this study aimed to assess and determine water quality using physicochemical parameters components for the Roxas City riverine communities. The physicochemical test will also serve as bases for the rehabilitation and development activities to restore and mitigate the threat of global warming and climate change. This further aimed to ascertain the following:

1. What is the level of Ph, Dissolved oxygen, Salinity, Conductivity, temperature, and Oxidation-Reduction Potential of two riverine communities of Roxas City?
2. What is the water quality level in the two riverine communities of Roxas City?
3. What could be the cause of the decline in water quality?
4. What awareness should the riverine communities implement to maintain the quality of rivers?

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(11) Methodology/ METHOD OF RESEARCH

(Water Sampling Procedure)

This study was anchored on the Panay River System Project. This study was conducted at the Palina River and Cadimahan River. Water samples of approximately 500 mL each were collected in 9 sections of the river. Sampling collection was performed one time at an interval of 30 days from March 2022 to October 2022. Water samples were collected at each point in the midstream, and the water depth of approximately 3-12 ft meters during the high tide period. Water samples were taken from the top, middle, and bottom, and both sides of the stream flow were mixed thoroughly. After sampling, the bottles were kept airtight and labeled correctly. A total of 2 gallons of water was collected from each sampling station using sampling bottles and transported in an ice chest for analytical test after collection. Using a portable device, the water was analyzed for Ph, Dissolved oxygen, Salinity, Conductivity, temperature, and Oxidation-Reduction Potential.

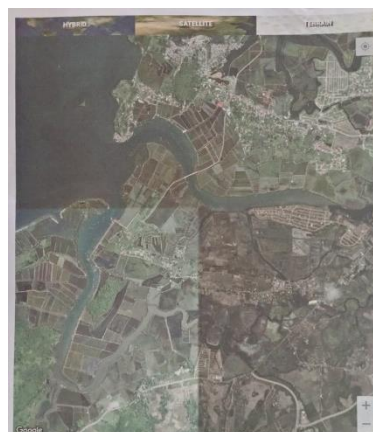


Fig. 1. Map of the study area and sampling sites in Palina River.

The Palina Greenbelt Ecopark is situated at Sitio Cablatan Brgy. Cagay has a longitude of 122° 42' 45.72" E and a latitude of 11° 34' 32.9232" N. In 2011, the primary operation was set in motion. The Palina Greenbelt Eco Park is one of the five eco-sites projects of the Roxas City Government. The Eco Park is well known for its floating boats or 'kabana' that tour around Cagay River's crystalline water, brimming with marine life. The floating boats can hold at least 15-20 people. It has a lot in store for visiting tourists; recreational fishing, sassing experience, foot spa massage, body massage, a wide array of food from the waters of Capiz, and a scenic view while dining.

The River Ecosystem in Palina is a biological system that resembles the conduits of the earth on the chance that we examine the satellite. The water comes from a high spot and then, at that point, down to its level, which we call the upstream and downstream. The estuarine region is known as the waterway passage, where fresh and saltwater meet. The water around here is called **brackish water**, which is considered bitter water with a pungent degree of 20-30 ppt. The water is prosperous in minerals due to the supplements that stream into the topsoil.

The Cadimahan River Tour, situated at Lawis Bay-Bay Roxas City, was the first river tour in Roxas City. It has a latitude of 29.3245331591 and a longitude of 47.9041405526. They are considered one of the major tourist attractions in Roxas City, Capiz, and are also known for promoting Eco-cultural tourism. The Cadimahan River passes through 7 barangays (i.e., Culasi, Libas, Bay-Bay, Tanque, Dayao), one of the biggest in all

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16 river systems in Capiz, as it spans around 68 hectares. The Cadimahan River Tour was conceptualized to provide alternative livelihood and promote Eco-cultural tourism. For the most part, communities and civilizations are developed around rivers since waterways have their navigational path, fishing ground, and water sources for the fish pond and households.



Fig. 2. Map of the study area and sampling sites in Cadimahan River.

(WATER QUALITY DETERMINATION)

In this paper, the water parameter measurements, the analytical methods, and the application of the Water Quality Index (WQI) to determine the physicochemical of the water samples were using APHA-AWWA (2005) methods of analysis (Table 1). Six (6) parameters were analyzed from water samples: Ph, Dissolved oxygen, Salinity, Conductivity, temperature, and Oxidation-Reduction Potential. The results were compared to the water quality standards for fresh surface waters such as lakes, rivers, reservoirs, etc., based on the DENR Administrative Order No. 34 s.1990.

Table 1. List of parameters and methods of analysis.

Parameter	Method of Analysis
Ph	Hach Portable multi-parameter metera
Dissolved oxygen	
Salinity	
Conductivity	
Temperature	
Oxidation-Reduction Potential	

Note aMeasured in-situ; Methods are based on Standards Methods of Analysis for Water and Wastewater, 21st Edition, APHA AWWA (2010)

The salinity of the water is referred to as the salt concentration in the water or land. Measuring the salinity in the water is very important in the lives of aquatic organisms and for agricultural purposes. Salinity is expressed in parts per thousand (ppt) or percentage (%). The method in determining the salinity in the Palina River and Cadimahan River water samples was based on the Department of Water, Government of Western Australia's standard, which the World Health Organization has acknowledged.

Salinity Status	Salinity (%)	Salinity (ppt)	Use
Fresh	< 0.05	< 0.5	Drinking and all irrigation
Marginal	0.05 – 0.1	0.5 – 1	Most irrigation, adverse effects on ecosystems become apparent
Brackish	0.1 – 0.2	1 – 2	Irrigation certain crops only; useful for most stock
Saline	0.2 – 1.0	2 – 10	Useful for most livestock
Highly Saline	1.0 – 3.5	10 – 35	Very saline groundwater, limited use for certain livestock
Brine	> 3.5	> 35	Seawater; some mining and industrial uses exist

Source: Department of Water. Government of Western Australia.

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Figure 1. Salinity of Water Determiner

Oxygen is a natural element needed by all forms of a living organisms. Almost all aquatic animals use dissolved oxygen in the water. It primarily enters the water through diffusion from the surrounding air and photosynthesis by aquatic plants. Dissolved oxygen is measured in units of milligrams/liter (mg/L) or as a percent of saturation (%). In determining the dissolved oxygen in the water samples of the Palina River and Cadimahan River, was based on the Application of Water Quality Indices and Dissolved Oxygen as Indicators for River Water Classification and Urban Impact Assessment.

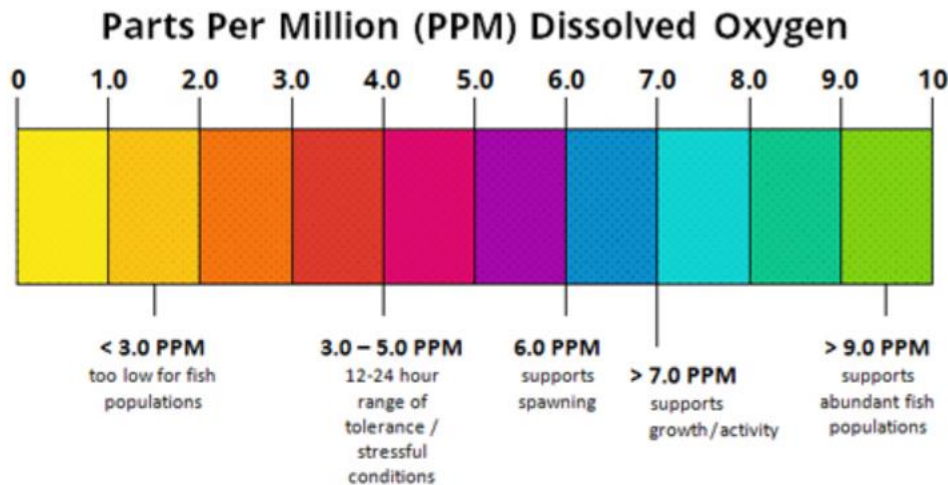
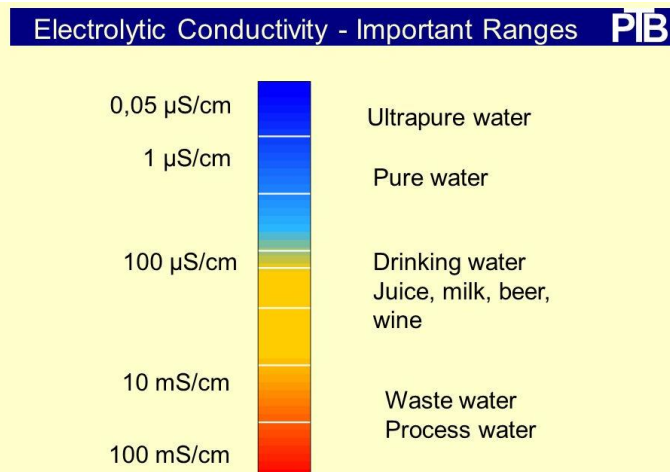


Figure 2. Range of Tolerance for Dissolved Oxygen

Conductivity is the ability of the water to conduct an electric current. It runs an electric current through the water between two metal electrodes. Conductivity assesses the health of a watercourse. The aquatic organism needs a constant concentration of dissolved ions in water. The conductivity of the water affects the growth and reproduction of marine organisms. Also, if the conductivity in the water decline, it signifies some pollution. The method for determining the conductivity in the Palina River and Cadimahan River water samples was based on the DataStream Initiative (2021), A Water Monitor's Quality.



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Figure 3. Conductivity Range Measurement

The ph in the water measures the alkalinity or the acidity in the water. It is the measurement of the activity of hydrogen atoms. The hydrogen atom is a good representation of the acidity and alkalinity of the water. The ph scale below (Figure 4) ranges from 0 to 14, with 7.0 being neutral. The higher the ph in the water indicates alkaline, while the lower ph indicates acidity. Pure water has a ph of 7.0. The method for determining the Ph in the Palina River and Cadimahan River water samples was based on the World Health Organization. 2007. pH in Drinking water: Revised background document for development of WHO Guidelines for Drinking-water Quality.

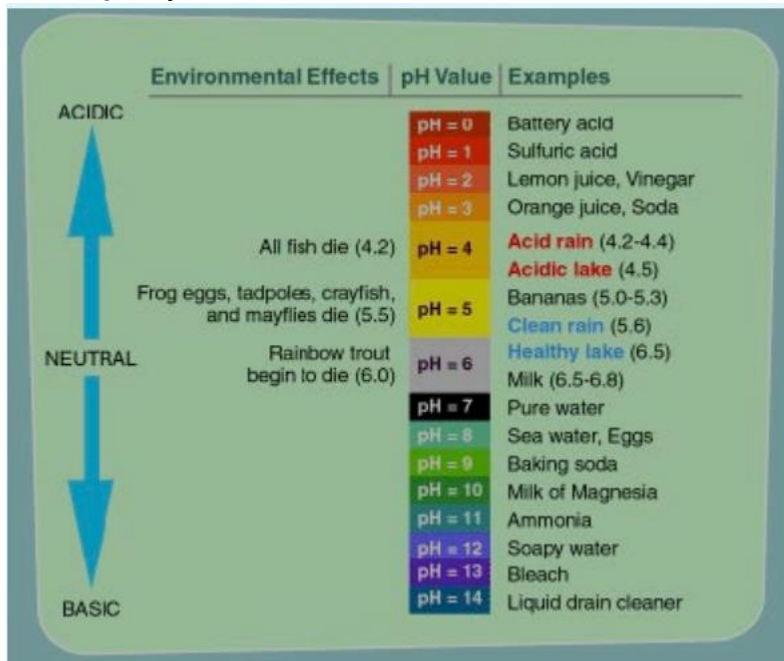


Figure 4. Ph Scale Meter

Water temperature is critical because it is an essential quality in environmental parameters. It is important to measure water temperature. By doing so, we can see the characteristics of the water, such as the chemical, biological, and physical properties of the water, as well as the possible health effects. Water temperature is essential in determining whether a body of water is acceptable for human consumption and use. Water temperature can affect metabolic rates and biological organisms. Organisms live in an optimal temperature range of 5-25 degrees Celsius. Water temperature should not exceed 89 degrees Fahrenheit.



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Figure 5.

Oxidation-reduction potential (ORP) measures the ability of a lake or river's ability to cleanse or break down waste products, such as contaminants, dead plants and animals. If the ORP in the water is high, it indicates abundant oxygen in the water. It signifies that the bacteria that decompose dead tissue can work efficiently. Generally, the ORP value in the water the healthier. ORP is also used to monitor drinking water and wastewater treatment. The method for determining the ORP in the water samples at Palina River and Cadimahan River was based on the standard of the World Health Organization.

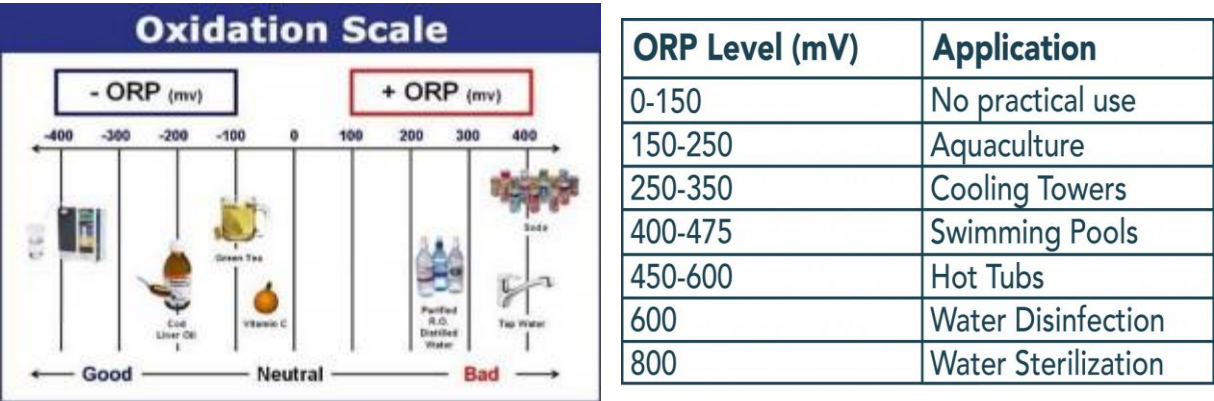


Figure 6. Oxidation-Reduction Potential (ORP) Scale

(12) Results and Discussions

On July 2016, the United Nations General Assembly (UN,2016) declared that access to clean and safe water consumption was one of the human rights and the protection of bodies of water. The physicochemical parameters of water are essential to determine the quality level of the water. The physicochemical parameters such as Ph, Dissolved oxygen, Salinity, Conductivity, temperature, and Oxidation-Reduction Potential were tested in the two riverine communities of Roxas City. Figure 7 shows the location of the water sampling with three



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sections.

Figure 7. Location of Station 1 and their sections

Phase 1: Water Quality Levels of Palina Greenbelt Eco Part (March 23, 2022)

The water samples were taken from three stations with three sections each. Sampling collection was conducted one time at an interval of 30 days from March 2022 to October 2022. Water samples were collected at each point in the midstream, and the water depth of approximately 3-12 ft meters during the high tide period. Below is the water analysis result obtained at Palina River on March 23, 2022.

Table 1: Physicochemical Analysis in Station 1 at Palina River

<i>Station 1</i>	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	3 ft	9.8	3.5	53.9 m/s	7.53	27 °C	339
<i>Middle</i>	7 ft	9.9	3.52	54.2 m/s	7.23	27 °C	340
<i>Left</i>	3 ft	9.9	3.49	53.9 m/s	7.14	27 °C	338

Station 1 Sampling Results

The Physico-chemical parameters obtained from the analysis of water samples were presented in Table 1. The water temperature value ranged from 27 °C; it was constant in three sections. Even though the three sections have different depths, the water temperature remains consistent. Water temperature is one of the aquatic ecosystems' most critical physical characteristics and affects several water quality parameters. The Ph value did not vary much from each section. Based on Figure 4. The ph Scale meter of the World Health

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Organization signifies that the Ph value in Station 1 was suitable for protecting aquatic habitats.

The conductivity value in station 1, in three sections, did not vary much. The conductivity in the right section, with a depth of 3 ft, was 53.9 m/s; in the middle area, with a depth of 7 ft, it was 54.2 m/s, and in the left section, the same depth in the left section was 53.9 m/s. Based on Figure 3, the Conductivity range measurement signifies that the water in Palina river is suitable for aquatic life.

Table 1 in station 1 shows the dissolved oxygen concentration in the three sections. The right section with a depth of 3 ft has a DO of 9.8 ppm, the middle section with a depth of 7 ft has a DO of 9.0 ppm, and the left section with a depth of 3 ft has a of DO 9.9 ppm. This signifies that the water in the Palina River has an abundant DO that supports the growth/population of marine life.

Based on Table 1 Oxidation-reduction potential (ORP) in the right section, has an ORP of 339 mV, in the middle section, a depth of 7 f,t it has an ORP of 340mV, and in the left section with a depth of 3 ft, it has an ORP of 338mV. The ORP in section 1 reaches above 300mV, which signifies that the water in section 1 is healthy.

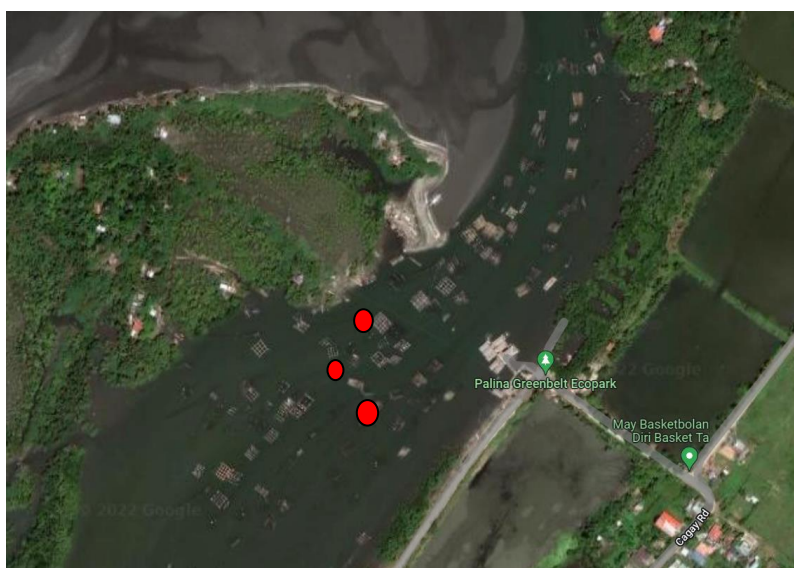


Figure 8. Location of Station 2 and their sections

Table 2: Physicochemical Analysis in Station 2 at Palina River

<i>Station 2</i>	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	3 ft	9.6	3.45	53.8 m/s	7.65	28 °C	372
<i>Middle</i>	7 ft	9.9	3.5	54.3 m/s	7.51	28 °C	372
<i>Left</i>	4 ft	9.9	3.47	54 m/s	7.3	28 °C	372

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Station 2 Sampling Results

Based on Table 2, the right, middle and left section has the exact temperature of 28 degrees Celsius even though has a different depth. It is slightly higher than Station 1; this this also signifies that marine life can thrive in this temperature. In the right section, it has a ph of; in5, in the middle section it has a 7.51 ph, and in the left section has a 7.3 ph. The ph value has different values because there were samples in different depths. The ORP in station 2 has remained constant in three sections, with different depths. The DO in station 2 indicates that the water in the left, middle, and right sections was abundant in oxygen that can efficiently support aquatic life. The three sections have a conductivity of 53.8 m/s, 54.3 m/s, and 54 m/s. Conductivity affects the survival of marine life. The conductivity at station 2 was desirable for the growth of marine organisms. The Palina River is a brackish water. It has more conductive than fresh water. The salinity in station 1 and station 2 has a close range in value.

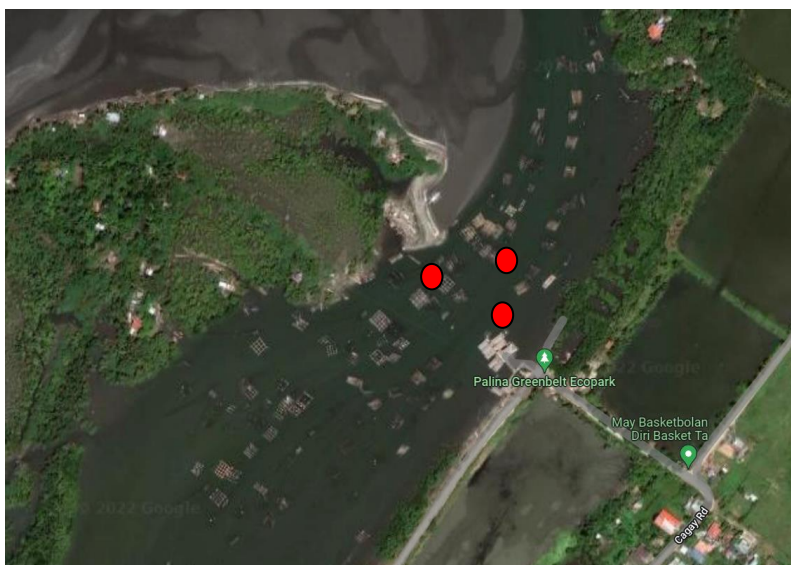


Figure 9. Location of the Station 3 and their sections

Table 3: Physicochemical Analysis in Station 3 at Palina River

Station 3	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
Right	3 ft	9.7	3.47	54.2 m/s	7.68	27.1 °C	329
Middle	7 ft	9.8	3.54	54.9 m/s	7.14	27.1 °C	376
Left	3 ft	9.9	3.48	54.2 m/s	7.36	27.1 °C	378

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Station 3 Sampling Results

Table 3 shows the result of water samples in the Palina River. The temperature in the three sections remains constant. In the right section, it has an ORP of 329 mV, in the middle area, it has 376 mV, and in the left section, it has 378 mV. The ORP in section 3 indicates that water samples in Palina were within the desirable limits based on the standard in Figure 6. Station 3 has a conductivity of 54.2 m/s, 54.9 m/s, and 54.2 m/s, this signifies that the water in station 3 has good conductivity. The right section has a DO of 9.7 ppm, in the middle section it has 9.8 ppm, and in the left section, it has 9.9 ppm. This indicates that the DO in section 3 is abundant that supports marine life. Section 3 has 7.68 ph, 7.14 ph, and 7.36 ph. The ph indicates that the water has an abundance of alkalinity. The salinity in station 3 has almost the same value in all the station in Palina.

Phase 1: Water Quality Levels of Cadimahan River (March 23, 2022)

The Cadimahan River Tour, situated at Lawis Bay-Bay Roxas City, was the first river tour in Roxas City. It has a latitude of 29.3245331591 and a longitude of 47.9041405526. The researcher took water samples from the three stations with three sections each. The waters collected were subjected to testing. The dissolved Oxygen, salinity, conductivity, Ph, temperature, and ORP were used to test the water samples of the Cadimahan River.

Table 4: Physicochemical Analysis in Station at Cadimahan River

<i>Station</i> 1	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	6 ft	9.9	3.33	52.9 m/s	7.36	27.7 °C	298
<i>Middle</i>	12 ft	9.8	3.30	52.8m/s	7.50	27.7 °C	295
<i>Left</i>	6 ft	9.9	3.30	52.8 m/s	7.65	27.7 °C	293

Station 1 Sampling Results

Table 4 shows the result of the water analysis of the Cadimahan River in station 1. In the right section, it has a depth of 6 ft; in the middle section, it has a 12 ft depth, and in the left section, it has a 6 ft depth. Based on the table above, the dissolved oxygen of 9.9 ppm remains constant in the left and right sections, while the middle section has 9.8 ppm. This indicates that the water in Cadimahan has an abundant DO that can support the growth of aquatic plants. Cadimahan River is a brackish type of water. The salinity ranges from 3.30 up to 3.33 ppt. The middle and left sections have the same conductivity of 52.8 m/s, and the right sections 52.9 m/s. The Ph value in station 1 ranges from 7.36-7.67. The Ph value differs in the depth of each section. The temperature remains constant in the three sections. The ORP value ranges vary from the different depths of the station.

Table 5: Physicochemical Analysis in Station at Cadimahan River

<i>Station</i> 2	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
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<i>Right</i>	6 ft	9.9	3.32	52.3m/s	7.61	28.0 °C	278
<i>Middle</i>	12 ft	9.8	3.29	51.9 m/s	7.55	28.0 °C	273
<i>Left</i>	6 ft	9.9	3.31	52.9 m/s	7.65	28.0 °C	254

Station 2 Sampling Results

In sampling station 2, the depth is still the same. The dissolved oxygen was similar to the result of station 1. The salinity in station 2 was slightly lower than in station 1, ranging from 3.29-3.32. the conductivity value ranges from 51.9 m/s-52.9 m/s, it has a lower result than station 1. The ph value in station 2 has a higher value range than station 1, ranging from 7.55-7.65pHs. The temperature in station 2 remains constant. The ORP in station 2 has a lower value than station 1.

Table 5: Physicochemical Analysis in Station at Cadimahan River

<i>Station</i> 3	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	6 ft	9.9	3.29	51.9 m/s	7.65	27.9 °C	252
<i>Middle</i>	12 ft	9.8	3.32	52.9 m/s	7.69	27.9 °C	252
<i>Left</i>	6 ft	9.9	3.29	52.1 m/s	7.65	27.9 °C	250

Station 3 Sampling Results

Based on table 5 above, the result shows that dissolved oxygen was still the same in stations 1, 2, and 3. The salinity level in station 3 was slightly lower than in station 2. The conductivity ranges from 51.9 m/s-52.9 m/s; stations 2 and 3 have the same conductivity range. The ph level ranges from 7.65-7.69, slightly higher than station 2. The temperature in station 3 was consistent depending, on the depth in each section. The ORP in station 3 was the lowest compared to stations 1 and 2.

Phase 2: Water Quality Levels of Palina Greenbelt Eco Part (April 25, 2022)

The researchers took another water sample at Palina River in the same three sections with three sections each. There is a one-month interval in collecting water samples. The collected water samples were subjected to testing with the same water parameters. Below is the water data analysis for the month of April 2022. The researcher took the same depth in all nine sections of the Palina River.

Table 6: Physicochemical Analysis in Station at Palina River

<i>Station</i> 1	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	3 ft	9.7	3.47	53.7 m/s	7.73	27.3 °C	329
<i>Middle</i>	7 ft	10	3.54	54.3 m/s	7.14	27.3 °C	385
<i>Left</i>	3 ft	9.9	3.41	53.9 m/s	7.4	27.3 °C	338

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Station 1 Sampling Results

Based on table 6 above, the dissolved oxygen in station 1 was higher than in last month's data test. The salinity becomes slightly higher ranging from 3.41-3.54 ppt, this indicates that the flow of water affects the salinity of the water. The conductivity remains constant, ranging from 53.7 m/s-54.3 m/s. The Ph level in station 1 was slightly higher compared to the first test. The temperature remains the same at 27 degrees Celsius. The ORP was still within the desirable limit.

Table 7: Physicochemical Analysis in Station at Palina River

Station 2	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
<i>Right</i>	3 ft	9.2	3.42	53.7 m/s	7.83	28.32 °C	372
<i>Middle</i>	7 ft	9.9	3.43	53.9 m/s	7.63	28.32 °C	372
<i>Left</i>	4 ft	9.9	3.45	54 m/s	7.28	28.32 °C	372

Station 2 Sampling Results

Table 7 shows the data on station 2 of the Palina River. The dissolved oxygen remains constant after one month of testing. The salinity in station 2 becomes lower. The conductivity remains the same. The Ph level increases by .32%, signifying that the water is more alkaline than in the previous test. The temperature was still the same, and no change. The ORP value has also increased, indicating that the Palina river has the ability to cleanse itself and break down waste products.

Table 8: Physicochemical Analysis in Station at Palina River

Station 3	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
<i>Right</i>	3 ft	9.7	3.47	53.7 m/s	7.73	27.3 °C	329
<i>Middle</i>	7 ft	10	3.54	54.3 m/s	7.14	27.3 °C	385
<i>Left</i>	3 ft	9.9	3.41	53.9 m/s	7.4	27.3 °C	338

Station 3 Sampling Results

Table 8 shows the data on station 3 of the Palina River. The dissolved oxygen remains constant after one month of testing. The salinity in station 3 becomes slightly higher than in station 2. The conductivity remains the same. The Ph level increases by .2%, signifying that the water is more alkaline than in the previous test. The temperature was still the same, and no change. The ORP value has also increased, indicating that the Palina river has the ability

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to cleanse itself and break down waste products.

Phase 2: Water Quality Levels of Cadimahan River (April 25, 2022)

The researcher took water samples at Cadimahan River in one-month intervals. Water samples were tested with the same parameters of water. The Cadimahan River Tour, situated at Lawis Bay-Bay Roxas City, was the first river tour in Roxas City. It has a latitude of 29.3245331591 and a longitude of 47.9041405526. The researcher took water samples from the three stations with three sections each. The waters collected were subjected to testing. The dissolved Oxygen, salinity, conductivity, Ph, temperature, and ORP were used to test the water samples of the Cadimahan River.

Table 9: Physicochemical Analysis in Station at Cadimahan River

Station 1	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
<i>Right</i>	6 ft	10	3.31	52.1 m/s	7.14	27.8 °C	296
<i>Middle</i>	12 ft	10	3.33	52.2 m/s	7.57	27.8 °C	288
<i>Left</i>	6 ft	10	3.29	51.7 m/s	7.65	27.8 °C	283

Station 1 Sampling Results

Table 9 shows the result of the water analysis of the Cadimahan River in station 1. In the right section, it has a depth of 6 ft; in the middle section, it has a 12 ft depth, and in the left section, it has a 6 ft depth. Based on the table above, the dissolved oxygen of 10 ppm remains constant in the three sections. This indicates that the water in Cadimahan has an abundant DO that can support the growth of aquatic plants. Cadimahan River is a brackish type of water. The salinity ranges from 3.29 up to 3.33 ppt. The middle and left sections have the same conductivity of 52.2 m/s, and the right sections are 52.2 m/s. The Ph value in station 1 ranges from 7.14-7.65. The Ph value differs in the depth of each section. The temperature remains constant in the three sections. The ORP value ranges vary from the different depths of the station.

Table 10: Physicochemical Analysis in Station at Cadimahan River

Station 2	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
<i>Right</i>	6 ft	9.8	3.29	52 m/s	7.71	28.2 °C	268
<i>Middle</i>	12 ft	10	3.28	51.6 m/s	7.75	28.3 °C	263
<i>Left</i>	6 ft	9	3.3	52.1 m/s	7.75	28.3 °C	244

Station 2 Sampling Results

Table 10 shows the result of the water analysis of the Cadimahan River in station 2. In the right section, it has a depth of 6 ft; in the middle section, it has a 12 ft depth, and in the left

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section, it has a 6 ft depth. Based on the table above, the dissolved oxygen of right section was 9.8 ppm, 10 ppm in the middle section and 9 ppm in the left section. This indicates that the water in Cadimahan has an abundant DO that can support the growth of aquatic plants. The salinity ranges from 3.30 up to 3.33 ppt. The middle and left sections have the same conductivity of 52.8 m/s, and the right sections 52.9 m/s. The Ph value in station 1 ranges from 7.71-7.75. The Ph value differs in the depth of each section. The temperature remains constant in the three sections. The ORP value ranges vary from the different depths of the station.

Table 11: Physicochemical Analysis in Station at Cadimahan River

Station 3	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
Right	3 ft	9.7	3.3	51.9 m/s	7.75	27.9 °C	242
Middle	7 ft	10	3.34	52.9 m/s	7.79	27.9 °C	242
Left	3 ft	9.9	3.29	52.1 m/s	7.7	27.9 °C	240

Station 3 Sampling Results

Table 11 shows the result of the water analysis of the Cadimahan River in station 3. Based on the table above, the dissolved oxygen in the right section was 9.7 ppm, 10 ppm in the middle section, and 9.9 ppm in the left section. This indicates that the water in Cadimahan has an abundant DO that can support the growth of aquatic plants. Station 1 and 2 have the same salinity value. The salinity ranges from 3.29 up to 3.33 ppt. The middle and left sections have the same conductivity of 52.8 m/s, and the right sections 52.9 m/s. The Ph value in station 1 ranges from 7.36-7.67. The Ph value differs in the depth of each section. The temperature remains constant in the three sections. The ORP value ranges vary from the different depths of the station.

Phase 3: Water Quality Levels of Palina River (May 23, 2022)

The Palina Greenbelt Ecopark is situated at Sitio Cablatan Brgy. Cagay has a longitude of 122° 42' 45.72" E and a latitude of 11° 34' 32.9232" N. In 2011, the primary operation was set in motion. The Palina Greenbelt Eco Park is one of the five eco-sites projects of the Roxas City Government. Collecting water samples every month was a continuous process.

Table 12: Physicochemical Analysis in Station at Palina River

Station 1	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
Right	3 ft	10	3.5	54.2 m/s	7.68	27.3 °C	329
Middle	7 ft	10	3.53	54.3 m/s	7.14	27.3 °C	385
Left	3 ft	9.8	3.43	53.8 m/s	7.4	27.3 °C	338

Station 1 Sampling Results

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Table 12 shows the data on station 1 of the Palina River. The dissolved oxygen in the right and middle section became higher after two months of testing, while the left section remains the same. The salinity in station 1 remains the same. The conductivity remains the same. The Ph level increases by .2%, signifying that the water is more alkaline than in the previous test. The temperature was still the same, and no change. The ORP value has also increased, indicating that the Palina river has the ability to cleanse itself and break down waste products.

Table 13: Physicochemical Analysis in Station at Palina River

<i>Station 2</i>	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	3 ft	9.9	3.44	53.6 m/s	7.83	28.32 °C	372
<i>Middle</i>	7 ft	9.8	3.43	53.7 m/s	7.63	28.32 °C	372
<i>Left</i>	4 ft	9.9	3.45	53.9 m/s	7.28	28.32 °C	372

Station 2 Sampling Results

Table 13 shows the data on station 2 of the Palina River. The dissolved oxygen remains constant after one month of testing. The salinity in station 2 becomes lower. The conductivity remains the same. The Ph level increases by .32%, signifying that the water is more alkaline than in the previous test. The temperature was still the same, and no change. The ORP value has also increased, indicating that the Palina river has the ability to cleanse itself and break down waste products.

Table 14: Physicochemical Analysis in Station at Palina River

<i>Station 3</i>	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	3 ft	9.9	3.47	53.7 m/s	7.73	27.3 °C	329
<i>Middle</i>	7 ft	10	3.54	54.3 m/s	7.14	27.3 °C	385
<i>Left</i>	3 ft	9.9	3.42	53.9 m/s	7.4	27.3 °C	338

Station 3 Sampling Results

Table 14 shows the data on station 2 of the Palina River. The dissolved oxygen remains constant after one month of testing. The salinity in station 2 becomes lower. The conductivity remains the same. The Ph level remains the same, signifying that the water is more alkaline than in the previous test. The temperature was still the same, and no change. The ORP value has also increased, indicating that the Palina river has the ability to cleanse itself and break down waste products.

Phase 3: Water Quality Levels of Cadimahan River (May 23, 2022)

The Cadimahan River Tour, situated at Lawis Bay-Bay Roxas City, was the first river tour in

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Roxas City. It has a latitude of 29.3245331591 and a longitude of 47.9041405526. The researcher took water samples from the three stations with three sections each. The waters collected were subjected to testing. The dissolved Oxygen, salinity, conductivity, Ph, temperature, and ORP were used to test the water samples of the Cadimahan River.

Table 15: Physicochemical Analysis in Station at Cadimahan River

<i>Station</i> 1	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	6 ft	10	3.31	52.1 m/s	7.14	27.8 °C	296
<i>Middle</i>	12 ft	10	3.33	52.1 m/s	7.57	27.8 °C	288
<i>Left</i>	6 ft	10	3.29	51.7 m/s	7.65	27.8 °C	283

Station 1 Sampling Results

Table 15 shows the result of the water analysis of the Cadimahan River in station 1. Based on the table above, the dissolved oxygen in the right, middle and left sections was 10 ppm, slightly higher than the previous test. This indicates that the water in Cadimahan has an abundant DO that can support the growth of aquatic plants. The salinity has the same value. The salinity ranges from 3.29 up to 3.33 ppt. The middle and right sections have the same conductivity of 52.1 m/s, and the left sections 51.7 m/s. The Ph value in station 1 ranges from 7.14-7.65. The Ph value differs in the depth of each section. The temperature remains constant in the three sections. The ORP value ranges vary from the different depths of the station.

Table 16: Physicochemical Analysis in Station at Cadimahan River

<i>Station</i> 2	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	6 ft	9.8	3.29	52 m/s	7.71	28.2 °C	268
<i>Middle</i>	12 ft	10	3.28	51.6 m/s	7.75	28.3 °C	263
<i>Left</i>	6 ft	9	3.3	52.1 m/s	7.75	28.3 °C	244

Station 2 Sampling Results

Table 16 shows the result of the water analysis of the Cadimahan River in station 2. Based on the table above, the dissolved oxygen in the right, middle and left sections was 10 ppm, slightly higher than the previous test. This indicates that the water in Cadimahan has an abundant DO that can support the growth of aquatic plants. The salinity has the same value. The salinity ranges from 3.29 up to 3.33 ppt. The middle and right sections have the same conductivity of 52.1 m/s, and the left sections 51.7 m/s. The Ph value in station 1 ranges from 7.14-7.65. The Ph value differs in the depth of each section. The temperature remains constant in the three sections. Even though the ORP value ranges vary from the different depths of the station it is still within the desirable limits.

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Table 17: Physicochemical Analysis in Station at Cadimahan River

Station 3	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
Right	3 ft	9.7	3.3	51.9 m/s	7.75	27.9 °C	242
Middle	7 ft	10	3.34	52.9 m/s	7.79	27.9 °C	242
Left	3 ft	9.9	3.29	52.1 m/s	7.7	27.9 °C	240

Station 3 Sampling Results

Table 17 shows the result of the water analysis of the Cadimahan River in station 3. The left section with a depth of 3 ft has a DO of 9.7 ppm, in the middle section with a depth of 7 ft has a DO of 10 ppm, and the left section with a depth of 3 ft has a DO of 9.9 ppm. Based on the analysis of the DO in section 3, Cadimahan River has abundant oxygen. The salinity, temperature, ORP, Ph, and conductivity in station 3 did not vary much from the previous data.

Phase 4: Water Quality Levels of Palina River (June 23, 2022)

On June 23, 2022, the researcher continues to collect water samples at the same point in the Palina River. These water samples were gathered and subjected to testing. The same water parameters were used in testing the water samples. The month of June was the 4th time of testing the water.

Table 18: Physicochemical Analysis in Station at Palina River

Station 1	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
Right	3 ft	9.9	3.44	53.6 m/s	7.83	28.32 °C	372
Middle	7 ft	9.8	3.43	53.7 m/s	7.63	28.32 °C	372
Left	4 ft	9.9	3.45	53.9 m/s	7.28	28.32 °C	372

Station 1 Sampling Results

Table 18 shows the result of the water analysis of the Palina River in station 1. Based on the data of water analysis in table 18, the dissolved oxygen, salinity, conductivity, ph, temperature, and ORP did not vary much from previous data results. This signifies that the fisherfolks of Palina River give importance to the value of the river and the river itself is abundant with marine life.

Table 19: Physicochemical Analysis in Station at Palina River

Station 2	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
Right	3 ft	9.9	3.47	53.7 m/s	7.73	27.3 °C	329

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<i>Middle</i>	7 ft	10	3.54	54.3 m/s	7.14	27.3 °C	385
	3 ft	9.9	3.42	53.9 m/s	7.4	27.3 °C	338

Station 2 Sampling Results

Table 19 shows the result of the water analysis of the Palina River in station 2. Based on the data of water analysis in table 19, the dissolved oxygen, salinity, conductivity, ph, temperature, and ORP did not vary much from previous data results. The result shows that after four times testing the water samples at Palina river, it has consistent data results.

Table 20: Physicochemical Analysis in Station at Palina River

<i>Station</i> 3	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	3 ft	10	3.5	54.2 m/s	7.68	27.3 °C	329
<i>Middle</i>	7 ft	10	3.53	54.3 m/s	7.14	27.3 °C	385
<i>Left</i>	3 ft	9.8	3.43	53.8 m/s	7.4	27.3 °C	338

Station 3 Sampling Results

Table 20 shows the result of the water analysis of the Palina River in station 3. Based on the data of water analysis in table 19, the dissolved oxygen, salinity, conductivity, ph, temperature, and ORP did not vary much from previous data results.

Phase 4: Water Quality Levels of Cadimahan River (June 23, 2022)

The researcher continues to collect water samples at the same point in Cadimahan River. These water samples were gathered and subjected to testing. The same water parameters were used in testing the water samples. The month of June was the 4th time testing the water.

Table 21: Physicochemical Analysis in Station at Cadimahan River

<i>Station</i> 1	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	6 ft	9.9	3.29	51.9 m/s	7.65	27.9 °C	252
<i>Middle</i>	12 ft	9.8	3.32	52.9 m/s	7.69	27.9 °C	252
<i>Left</i>	6 ft	9.9	3.29	52.1 m/s	7.65	27.9 °C	250

Station 1 Sampling Results

Table 20 shows the result of the water analysis of the Cadimahan River in station 1. Based on the data of water analysis in table 21, the dissolved oxygen, salinity, conductivity, ph, temperature, and ORP did not vary much from previous data results. The result shows that after the fourth time of testing the Cadimahan River the data indicated the same consistent

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result.

Table 22: Physicochemical Analysis in Station at Cadimahan River

Station 2	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
<i>Right</i>	6 ft	9.9	3.29	51.9 m/s	7.65	27.9 °C	252
<i>Middle</i>	12 ft	9.8	3.32	52.9 m/s	7.69	27.9 °C	252
<i>Left</i>	6 ft	9.9	3.29	52.1 m/s	7.65	27.9 °C	250

Station 2 Sampling Results

Table 20 shows the result of the water analysis of the Cadimahan River in station 3. Based on the data of water analysis in table 22, the dissolved oxygen, salinity, conductivity, ph, temperature, and ORP did not vary much from previous data results. The Cadimahan river is one of the largest rivers in the riverine communities of Roxas City, this also signifies that the fisherfolks who relied solely on their income on the river know how to properly take care of the river. The river has a lot to offer the riverine communities.

Table 23: Physicochemical Analysis in Station at Cadimahan River

Station 3	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
<i>Right</i>	6 ft	9.9	3.29	51.9 m/s	7.65	27.9 °C	252
<i>Middle</i>	12 ft	9.8	3.32	52.9 m/s	7.69	27.9 °C	252
<i>Left</i>	6 ft	9.9	3.29	52.1 m/s	7.65	27.9 °C	250

Station 3 Sampling Results

Table 23 shows the result of the water analysis of the Cadimahan River in station 3. Based on the data of water analysis in table 23, the dissolved oxygen, salinity, conductivity, ph, temperature, and ORP did not vary much from previous data results. The result shows that after the fourth time of testing the Cadimahan River the data indicated the same consistent result.

Phase 5: Water Quality Levels of Palina River (July 23, 2022)

The researcher took water samples on July 23rd, 2022. The water samples that were collected were subjected to water analysis. During high tide, the same section was used in collecting the water samples. The collected water samples were transported for testing after collection.

Table 24: Physicochemical Analysis in Station at Palina River

Station 1	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
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<i>Right</i>	3 ft	9.7	3.47	53.7 m/s	7.73	27.3 °C	329
<i>Middle</i>	7 ft	10	3.54	54.3 m/s	7.14	27.3 °C	385
<i>Left</i>	3 ft	9.9	3.41	53.9 m/s	7.4	27.3 °C	338

Station 1 Sampling Results

Table 24 shows the result of the water analysis of the Palina River in station 1. Based on the collected water samples, the dissolved oxygen, salinity, conductivity, ph, temperature, and ORP did not vary much from previous data results. The result signifies that the water in Palina river is suitable for aquatic life.

Table 25: Physicochemical Analysis in Station at Palina River

<i>Station 2</i>	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	3 ft	9.2	3.42	53.7 m/s	7.83	28.32 °C	372
<i>Middle</i>	7 ft	9.9	3.43	53.9 m/s	7.63	28.32 °C	372
<i>Left</i>	4 ft	9.9	3.45	54 m/s	7.28	28.32 °C	372

Station 2 Sampling Results

Table 25 shows the result of the water analysis of the Palina River in station 2. Based on the collected water samples, the dissolved oxygen, salinity, conductivity, ph, temperature, and ORP did not vary much from previous data results. The fifth test of the water samples shows that the water in the Palina river can sustain the growth of the aquatic organism.

Table 26: Physicochemical Analysis in Station at Palina River

<i>Station 3</i>	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	3 ft	9.7	3.47	53.7 m/s	7.73	27.3 °C	329
<i>Middle</i>	7 ft	10	3.54	54.3 m/s	7.14	27.3 °C	385
<i>Left</i>	3 ft	9.9	3.41	53.9 m/s	7.4	27.3 °C	338

Station 3 Sampling Results

Table 26 shows the result of the water analysis of the Palina River in station 2. Based on the collected water samples, the dissolved oxygen, salinity, conductivity, ph, temperature, and ORP did not vary much from previous data results. The fifth test of the water samples shows that the water in the Palina river can sustain the growth of the aquatic organism.

Phase 5: Water Quality Levels of Cadimahan River (July 23, 2022)

The researcher continues to collect water samples at the same point in Cadimahan River. These water samples were gathered and subjected to testing. The same water parameters

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were used in testing the water samples. The month of July was the 5th time testing the water. The water samples were collected during high tide.

Table 27: Physicochemical Analysis in Station at Cadimahan River

<i>Station</i> <i>1</i>	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	6 ft	9.9	3.29	51.9 m/s	7.65	27.9 °C	252
<i>Middle</i>	12 ft	9.8	3.32	52.9 m/s	7.69	27.9 °C	252
<i>Left</i>	6 ft	9.9	3.29	52.1 m/s	7.65	27.9 °C	250

Station 1 Sampling Results

Table 27 shows the result of the water analysis of the cadimahan River in station 1. The same depth was in station 1 where the water samples were collected. The dissolved oxygen ranges from 9.8-9.9 ppm, which signifies the abundance of oxygen in the water. The salinity ranges from 3.29-3.32 ppt, indicating that the water has enough salt for the growth of aquatic organisms. The conductivity value from 51.9 m/s- 52.9 m/s, based on the standard on figure 3, is within a desirable limit. The Ph level, the temperature, and the ORP has the same on the previous data collected.

Table 28: Physicochemical Analysis in Station at Cadimahan River

<i>Station</i> <i>2</i>	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	6 ft	9.9	3.29	51.9 m/s	7.65	27.9 °C	252
<i>Middle</i>	12 ft	9.8	3.32	52.9 m/s	7.69	27.9 °C	252
<i>Left</i>	6 ft	9.9	3.29	52.1 m/s	7.65	27.9 °C	250

Station 2 Sampling Results

Table 28 shows the result of the water analysis of the Cadimahan River in station 2. The same depth was in station 2 where the water samples were collected. The dissolved oxygen ranges from 9.8-9.9 ppm, which signifies the abundance of oxygen in the water. The salinity ranges from 3.29-3.32 ppt, indicating that the water has enough salt for the growth of aquatic organisms. The conductivity value from 51.9 m/s- 52.9 m/s, based on the standard on figure 3, is within a desirable limit. The Ph level, the temperature, and the ORP have the same as the previous data collected.

Table 29: Physicochemical Analysis in Station at Cadimahan River

<i>Station</i> <i>3</i>	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	6 ft	9.9	3.29	51.9 m/s	7.65	27.9 °C	252
<i>Middle</i>	12 ft	9.8	3.32	52.9 m/s	7.69	27.9 °C	252

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<i>Left</i>	6 ft	9.9	3.29	52.1 m/s	7.65	27.9 °C	250
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Station 3 Sampling Results

Table 29 shows the result of the water analysis of the Cadimahan River in station 3. The same depth was in station 1 where the water samples were collected. The dissolved oxygen ranges from 9.8-9.9 ppm, which was the same data in station 1. The salinity ranges from 3.29-3.32 ppt, the conductivity value from 51.9 m/s- 52.9 m/s, the ph level ranges from 7.65-7.69, the temperature of 27 degrees Celcius and the ORP value of 250-252 were the same in the data of station 1.

Phase 6: Water Quality Levels of Palina River (August 25, 2022)

On August 25th, 2022, the researcher continues to collect water samples at the same point in the Palina River. These water samples were gathered and subjected to testing. The same water parameters were used in testing the water samples. The month of August was the 6th time testing the water.

Table 30: Physicochemical Analysis in Station at Palina River

<i>Station 1</i>	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	3 ft	8.9	3.0	50.0 m/s	6.65	28.0 °C	252
<i>Middle</i>	7 ft	8.8	3.1	52.0 m/s	6.69	29.0 °C	252
<i>Left</i>	3 ft	8.5	3.1	51.0 m/s	6.65	30.0 °C	250

Station 1 Sampling Results

Table 30 shows the result of the water analysis of the Palina River in station 1. The same depth was in station 1 where the water samples were collected. The dissolved oxygen ranges from 8.5-8.9 ppm, the data shows that dissolved oxygen during the 6th test is lower compared to the previous test. The salinity ranges from 3.-3.1 ppt, showing a fluctuation of the salt in the water. The conductivity value from 50.0 m/s- 52.0 m/s, becomes lower compared to the previous test. The Ph level ranges from 6.65-6.69, a bit lower than the previous test. The temperature ranges from 28-30 degrees Celsius, and the ORP value of 250-252 was the same in the data of station 1.

Table 31: Physicochemical Analysis in Station at Palina River

<i>Station 2</i>	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	3 ft	8.8	3.0	50.0 m/s	6.0	27.0 °C	282
<i>Middle</i>	7 ft	8.5	3.1	52.0 m/s	6.51	27.50 °C	242

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<i>Left</i>	4 ft	8.5	3.1	51.0 m/s	6.43	26.60 °C	260
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Station 2 Sampling Results

Table 31 shows the result of the water analysis of the Palina River in station 2. The same depth was in station 1 where the water samples were collected. The dissolved oxygen ranges from 8.5-8.9 ppm, the data shows that stations 1 and 2 have the same dissolved oxygen. The salinity ranges from 3.-3.1 ppt, showing a fluctuation of the salt in the water. The conductivity value from 50.0 m/s- 52.0 m/s, becomes lower compared to the previous test. The Ph level ranges from 6.0-6.43, a bit lower than the previous test. The temperature ranges from 28-30 degrees Celsius, and the ORP value of 250-252.

Table 32: Physicochemical Analysis in Station at Palina River

<i>Station</i> 3	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	3 ft	8.8	3.0	51.9 m/s	6.0	30.0 °C	252
<i>Middle</i>	7 ft	8.5	3.1	52.9 m/s	6.51	27.50 °C	252
<i>Left</i>	4 ft	8.5	3.1	52.1 m/s	6.43	26.60 °C	250

Station 3 Sampling Results

Table 32 shows the result of the water analysis of the Palina River in station 3. The same depth was in station 1 where the water samples were collected. The dissolved oxygen ranges from 8.5-8.9 ppm, the data shows that stations 1, 2, and 3 have the same dissolved oxygen during the 6th test is lower compared to the previous test. The salinity in stations 1, 2, and 3 was the same in the 6th test of water samples. The conductivity in station 3 was a bit higher than stations 1 and 2. The Ph level becomes lower compared to the previous test. The conductivity value from 50.0 m/s- 52.0 m/s, was the same with stations 1 and 2. The temperature in the right section was 30 degrees Celsius, which is not suitable for some fish. The ORP value of 250-252 was the same in the data of station 1.

Phase 6: Water Quality Levels of Cadimahan River (August 25, 2022)

On August 25th, 2022, the researcher continues to collect water samples at the same point in the Cadimahan River. These water samples were gathered and subjected to testing. The same water parameters were used in testing the water samples. The month of August was the 6th time testing the water.

Table 33: Physicochemical Analysis in Station at Cadimahan River

<i>Station</i> 1	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	6 ft	7.10	3.0	51.9 m/s	7.0	29 °C	240
<i>Middle</i>	12 ft	6.36	3.1	52.9 m/s	7.2	29 °C	232

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<i>Left</i>	6 ft	7.10	3.1	52.1 m/s	7.5	29 °C	220
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Station 1 Sampling Results

Table 33 shows the result of the water analysis of the Cadimahan River in station 1. The data shows that the dissolved oxygen in the three sections becomes lower compared to the previous test conducted. The salinity, ORP, Ph, and conductivity becomes lower in the 6th time of testing the water samples. The temperature increases which is not suitable for some aquatic organisms.

Table 34: Physicochemical Analysis in Station at Cadimahan River

<i>Station 2</i>	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	6 ft	6.58	3.0	51.9 m/s	6.89	28 °C	252
<i>Middle</i>	12 ft	7.01	3.1	52.9 m/s	6.69	29 °C	252
<i>Left</i>	6 ft	6.10	3.1	52.1 m/s	6.65	29 °C	250

Station 2 Sampling Results

Table 34 shows the result of the water analysis of the Cadimahan River in station 2. The data shows that the dissolved oxygen in the three sections becomes lower compared to the previous test conducted. The salinity, ORP, Ph, and conductivity becomes lower in the 6th time of testing the water samples. The temperature increases which is not suitable for some aquatic organisms.

Table 35: Physicochemical Analysis in Station at Cadimahan River

<i>Station 3</i>	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	6 ft	6.91	3.0	51.9 m/s	7.5	29 °C	252
<i>Middle</i>	12 ft	7.30	3.1	52.9 m/s	7.6	29 °C	252
<i>Left</i>	6 ft	7.69	3.1	52.1 m/s	6.65	30 °C	250

Station 3 Sampling Results

Table 35 shows the result of the water analysis of the Cadimahan River in station 3. The data shows that the dissolved oxygen in the three sections becomes lower compared to the previous test conducted. The dissolved oxygen ranges from 6.92 ppm-7.69 ppm. The salinity, ORP, Ph, and conductivity becomes lower in the 6th time of testing the water samples. The temperature increases which is not suitable for some aquatic organisms.

Phase 7: Water Quality Levels of Palina River (September 25, 2022)

The researcher continues to collect water samples at the same point in the Palina River.

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These water samples were gathered and subjected to testing. The same water parameters were used in testing the water samples. The month of September was the 7th time testing the water. The water samples were collected during high tide.

Table 36: Physicochemical Analysis in Station at Palina River

Station 1	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
<i>Right</i>	3 ft	6.9	3.5	40.0 m/s	6.1	30.0 °C	222
<i>Middle</i>	7 ft	6.8	3.9	42.0 m/s	6.0	30.0 °C	222
<i>Left</i>	3 ft	6.5	3.9	41.0 m/s	6.1	30.0 °C	220

Station 1 Sampling Results

Table 35 shows the result of the water analysis of the Palina River in station 1. The data shows that the dissolved oxygen in the three sections becomes lower compared to the previous test conducted. The dissolved oxygen ranges from 6.92 ppm-7.69 ppm. The salinity, ORP, Ph, and conductivity becomes lower in the 6th time of testing the water samples. The temperature increases which is not suitable for some aquatic organisms.

Table 37: Physicochemical Analysis in Station at Palina River

Station 2	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
<i>Right</i>	3 ft	6.9	3.5	42.0 m/s	6.0	32.0 °C	232
<i>Middle</i>	7 ft	6.8	3.9	42.0 m/s	6.0	30.0 °C	225
<i>Left</i>	4 ft	6.5	3.9	41.0 m/s	6.0	31.0 °C	230

Station 2 Sampling Results

Table 37 shows the result of the water analysis of the Palina River in station 2. Based on the data above the dissolved oxygen ranges from 6.5-6.9 ppm in station 2 at Palina. The data shows the decline of dissolved oxygen in the water. The salinity level has increased which are not good for some aquatic organism. The conductivity decreases. The Ph level from station 2 in three sections was 6.0, which signifies that some of the aquatic organisms won't survive. The temperature becomes higher, and the ORP level decreases.

Table 38: Physicochemical Analysis in Station at Palina River

Station 3	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
<i>Right</i>	3 ft	6.9	3.9	40.0 m/s	6.1	31.0 °C	225
<i>Middle</i>	7 ft	6.8	3.7	40.0 m/s	6.0	32.0 °C	221
<i>Left</i>	4 ft	6.5	3.98	42.0 m/s	6.2	31.0 °C	225

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Station 3 Sampling Results

Table 38 shows the result of the water analysis of the Palina River in station 3. Based on the data above the dissolved oxygen ranges from 6.5-6.9 ppm in station 3 the same data result in station 2. The data shows the decline of dissolved oxygen in the water. The salinity level has increased which are not good for some aquatic organism. The conductivity was the same in stations 2 and 3. The Ph level from stations 2 and 3 in the three sections was 6.0, which signifies that some of the aquatic organisms won't survive. The temperature becomes higher, and the ORP level decreases.

Phase 7: Water Quality Levels of Cadmahan River (September 25, 2022)

The researcher continues to collect water samples at the same point at Cadimahan River. These water samples were gathered and subjected to testing. The same water parameters were used in testing the water samples. The month of September was the 7th time testing the water. The water samples were collected during high tide.

Table 39: Physicochemical Analysis in Station at Cadimahan River

Station 1	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
<i>Right</i>	6 ft	7.54	3.29	41.9 m/s	6.1	30 °C	252
<i>Middle</i>	12 ft	7.39	3.32	42.9 m/s	6.1	31 °C	252
<i>Left</i>	6 ft	7.39	3.29	42.1 m/s	6.2	31 °C	250

Station 1 Sampling Results

Table 39 shows the result of the water analysis of the Cadimahan River in station 1. Based on the data above the dissolved oxygen ranges from 7.29-7-32 ppm in station 1. The data shows the decline of dissolved oxygen in the water. The salinity level has increased which are not good for some aquatic organism. The conductivity has decreased compared to the last data result. The Ph level in station 1 decreased by 1%, which signifies that some of the aquatic organisms won't survive. The temperature becomes higher, and the ORP level decreases.

Table 40: Physicochemical Analysis in Station at Cadimahan River

Station 2	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
<i>Right</i>	6 ft	7.77	3.29	41.9 m/s	6.1	31 °C	252
<i>Middle</i>	12 ft	7.22	3.32	42.9 m/s	6.2	30 °C	252
<i>Left</i>	6 ft	7.20	3.29	42.1 m/s	6.2	30 °C	250

Station 2 Sampling Results

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Table 40 shows the result of the water analysis of the Cadimahan River in station 2. Based on the data above the dissolved oxygen ranges from 7.20-7.77 ppm in station 2. The data shows the decline of dissolved oxygen in the water. The salinity level has increased in stations 1 and 2, which are not good for some aquatic organisms. The conductivity has decreased compared to the last data result. The Ph level in stations 2 and 1 has the same level which signifies that some of the aquatic organisms won't survive. The temperature becomes higher, and the ORP level decreases.

Table 41: Physicochemical Analysis in Station at Cadimahan River

<i>Station</i> 3	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	6 ft	7.10	3.29	41.9 m/s	6.1	30 °C	252
<i>Middle</i>	12 ft	6.36	3.32	42.9 m/s	6.0	31 °C	252
<i>Left</i>	6 ft	7.10	3.29	42.1 m/s	6.0	30 °C	250

Station 3 Sampling Results

Table 41 shows the result of the water analysis of the Cadimahan River in station 3. Based on the data above, the dissolved oxygen levels vary according to their depth. The salinity, Ph, conductivity, temperature, and ORP level remain the same depending on the depth in each section.

Phase 8: Water Quality Levels of Palina River (October 20, 2022)

The researcher continues to collect water samples at the same point in the Palina River. These water samples were gathered and subjected to testing. The same water parameters were used in testing the water samples. The water samples were collected during high tide.

Table 42: Physicochemical Analysis in Station at Palina River

<i>Station</i> 1	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	3 ft	6.9	3.8	40.0 m/s	6.5	30.0 °C	210
<i>Middle</i>	7 ft	6.8	3.9	40.0 m/s	6.0	30.0 °C	222
<i>Left</i>	3 ft	6.5	3.1	41.0 m/s	6.1	30.0 °C	220

Station 1 Sampling Results

Table 42 shows the result of the water analysis of the Palina River in station 1. Based on the data above, the dissolved oxygen levels have the same data result on the last month's test. The salinity, Ph, conductivity, temperature, and ORP level has the same data result.

Table 43: Physicochemical Analysis in Station at Palina River

<i>Station</i>	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
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<i>Right</i>	3 ft	6.8	3.9	40.9 m/s	6.0	30.0 °C	232
<i>Middle</i>	7 ft	7.5	3.9	42.0 m/s	6.1	30.0 °C	242
<i>Left</i>	4 ft	7.5	4.0	41.0 m/s	6.3	30.0 °C	220

Station 2 Sampling Results

Table 43 shows the result of the water analysis of the Palina River in station 2. Based on the data above, the dissolved oxygen levels vary according to their depth. In the left section, it has a salinity level of 4ppt, indicating highly saline water. The conductivity in stations 1 and 2 has the same data result. The ph levels vary according to their depth. The temperature increases. The ORP level in stations 1 and 2 has the same data result.

Table 44: Physicochemical Analysis in Station at Palina River

<i>Station</i> 3	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	3 ft	7.8	3.9	41.9 m/s	6.0	30.0 °C	220
<i>Middle</i>	7 ft	6.5	3.19	42.9 m/s	6.1	30.0 °C	232
<i>Left</i>	4 ft	7.5	3.9	42.1 m/s	6.3	30.0 °C	240

Station 3 Sampling Results

Table 44 shows the result of the water analysis of the Palina River in station 3. Based on the data above, the dissolved oxygen levels vary according to their depth. In the right and left sections, it has a salinity level of 3.9ppt, indicating highly saline water. The conductivity in stations 1, 2, and 3 has the same data result. The ph levels vary according to their depth. The temperature increases. The ORP level in station 3 varies according to their depth.

Phase 8: Water Quality Levels of Cadimahan River (October 20, 2022)

The researcher continues to collect water samples at the same point in the Cadimahan River. These water samples were gathered and subjected to testing. The same water parameters were used in testing the water samples. The water samples were collected during high tide.

Table 45: Physicochemical Analysis in Station at Cadimahan River

<i>Station</i> 1	<i>Depth</i>	<i>DO</i>	<i>Salinity</i>	<i>Conductivity</i>	<i>PH</i>	<i>Temperature</i>	<i>ORP</i>
<i>Right</i>	6 ft	6.8	3.29	41.9 m/s	7.65	31 °C	252
<i>Middle</i>	12 ft	7.5	3.32	42.9 m/s	7.69	31 °C	252
<i>Left</i>	6 ft	7.5	3.29	42.1 m/s	7.65	31 °C	250

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Station 1 Sampling Results

Table 45 shows the result of the water analysis of the Cadimahan River in station 1. Based on the data above, the dissolved oxygen levels vary according to their depth. The data result of salinity and conductivity were the same as last month's data results. The ph level increased in station 1. The conductivity level in the middle section was higher compared to the left and right sections. The temperature increased to 31 degrees Celsius. The ORP level remains constant.

Table 46: Physicochemical Analysis in Station at Cadimahan River

Station 2	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
<i>Right</i>	6 ft	7.8	3.29	40.0 m/s	7.65	31 °C	252
<i>Middle</i>	12 ft	6.5	3.32	40.0 m/s	7.69	31 °C	252
<i>Left</i>	6 ft	7.5	3.29	41.0 m/s	7.65	31 °C	250

Station 1 Sampling Results

Table 46 shows the result of the water analysis of the Cadimahan River in station 2. Based on the data above, the dissolved oxygen levels vary according to their depth. The data result of salinity and conductivity were the same as last month's data results. The ph level increased in station 1. The conductivity level in the middle section was higher compared to the left and right sections. The temperature increased to 31 degrees Celsius. The ORP level remains constant.

Table 47: Physicochemical Analysis in Station at Cadimahan River

Station 3	Depth	DO	Salinity	Conductivity	PH	Temperature	ORP
<i>Right</i>	6 ft	6.8	3.9	40.9 m/s	7.65	31 °C	252
<i>Middle</i>	12 ft	7.5	3.2	42.0 m/s	7.69	31 °C	252
<i>Left</i>	6 ft	7.5	3.9	41.0 m/s	7.65	31 °C	250

Station 2 Sampling Results

Table 47 shows the result of the water analysis of the Cadimahan River in station 3. Based on the data above, the dissolved oxygen levels vary according to their depth. The data result of salinity and conductivity were the same in stations 1 and 2. The ph level increased in station 3 similar to station 2. The conductivity level in the middle section was higher compared to the left and right sections. The temperature increased to 31 degrees Celsius. The ORP level remains constant.

Water quality level in the two riverine communities of Roxas City

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The physicochemical parameters, such as temperature, pH, Dissolved Oxygen (DO), conductivity, salinity, and ORP were used in testing the water samples in the two riverine communities of Roxas City. These Physico-chemical parameters are used as indicators of ecosystem health and can be potentially linked directly to socio-economic indicators such as access to safe water, environmental or watershed protection expenditures, withdrawal of surface water, and domestic consumption of water per capita. These indicators and discussions could further be reinforced by providing a general description of the prevailing water quality of the rivers and the monitoring stations and can be done through the calculation of the water quality index.

The Physico-chemical parameters such as water temperature, conductivity, salinity (3.0ppt~4.0ppt), Ph, ORP, and dissolved oxygen(6.3ppm~10ppm) of both rivers were within desirable limits. The Dissolved Oxygen decreased 4ppm. The DO is 6.3 ppm; although it is near the borderline of a stressful condition for the fish population it is still acceptable. The salinity levels range from 3.0 ppt- 4.0 ppt. Based on the standard in Figure 1, the salinity level of the two rivers was within desirable limits, although there was a spike in salinity it is still tolerable for the aquatic organisms. The conductivity in the water measures the flow of electricity flowing in the current. The conductivity ranges from 40.0m/s~54.9m/s, based on figure 3, there was a decline in 10 m/s conductivity in the rivers. The ph level ranges from 6.0pH~7.83pH, which signifies that the two rivers have a desirable level of Ph. The standard water temperature for the aquatic organisms was 25-28 degrees Celsius. Based on the data collected by the researcher, the water temperature ranges from 26.60°C~28.32°C which is desirable for marine life. ORP is measured directly in the two rivers of Roxas City using an ORP sensor. ORP is measured in millivolts (mV) and the more oxygen that is present in the water, the higher the ORP reading is. The ORP ranges from 220 mV-385mV, which signifies that it is within the required limits. Based on the results acquired by the researcher water quality was good and did not show any significant pollution problem. The results from March to August were consistent in 9 sections of the two rivers. The results in September and October decreased. The decline of water results in the two rivers was due to red tide. According to National Geographic, there are at least three species of dinoflagellates, and one diatom species is responsible for red tides' toxic microscopic forms of algae that produce toxins that can sicken humans and be fatal for marine animals. The red tide affected both rivers, causing some fish to die.

Possible Causes of Water Decline in the River Communities

Rivers, lakes, and water reservoirs are the planet's essential freshwater resources that provide innumerable benefits. Rivers are used for irrigation purposes and provide an ecosystem for aquatic life. They have immense significant elements in the world's biological diversity. They have substantial social and economic benefits due to tourism and recreation and are culturally and aesthetically crucial for people worldwide. According to Appadurai (2001), one of the possible declines in water was the mussel and fish cages that hindered the water flow. Illegal fish pens, improper waste disposal, industrial and agricultural activity. The use of microplastic in the rivers,

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Awareness Campaign that Should be Implemented at the Riverine Communities

The result of the study implied that the fisherfolks are aware of how to take care of the river. The residents living along the Palina and Cadimahan Rivers sometimes received environmental education. Increase environmental education by the LGU, Department of Fisheries and Aquatic Resources, and Department of Environment Natural Resources; once all the residents are aware of the importance of rivers, the yearning for ecological restoration will not be far from reality.

(13) Conclusions and Recommendations

CONCLUSIONS

Based on the findings of this study, the following conclusions are drawn: The results obtained from the present study shall be useful in future research. By assessing the Physico-chemical data, the Palina river and Cadimahan river water quality were determined. Based on the values of the obtained Physico-chemical parameters, it can be concluded that the Palina and Cadimahan River water quality was good and did not show any significant pollution problem in the present study. Although, there were some points of irregularities during certain months of the year where the water quality was unbalanced.

RECOMMENDATIONS

Based on the drawn conclusions, the following recommendations were formulated:

1. Future researchers should test different water parameters such as total dissolved solids, total suspended solids, secchi disc depth, nitrate, nitrite, orto-phosphate, sulfate, chloride, total hardness, calcium, and magnesium.
2. Future researchers should conduct water tests in other riverine communities of Roxas City (Cagay, Barra, Libas, etc).
3. Acquire reliable portable tools for the testing of water.
4. Future researchers should re-conduct Physico-chemical analysis on a yearly basis in order to observe the long-term results of the water quality trend of the Capiz rivers.

(14) Output

An information and awareness campaign recorded video clip.

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

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(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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